

TrES-3b

R-0326

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170502 · created 2026-07-18T03:28:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457875.8871 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.022
Transit depth (Rp/Rs)^2	3.81 +/- 0.86 [%]
Orbital Inclination (inc)	81.52 +/- 0.53
Airmass coefficient 1 (a1)	1.117 +/- 0.013
Airmass coefficient 2 (a2)	-0.102 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	None
Optimal Aperture	3.87
Optimal Annulus	11.83
Transit Duration (day)	0.0554 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.022 vs 0.16309 (deviation 1.45σ)
Duration fit vs published	0.0554 d vs 0.0567 d (deviation 0.23σ)
Mid-time O–C	-1.2 min
Depth SNR	8.9
Residual scatter	1.17 %

Light curve

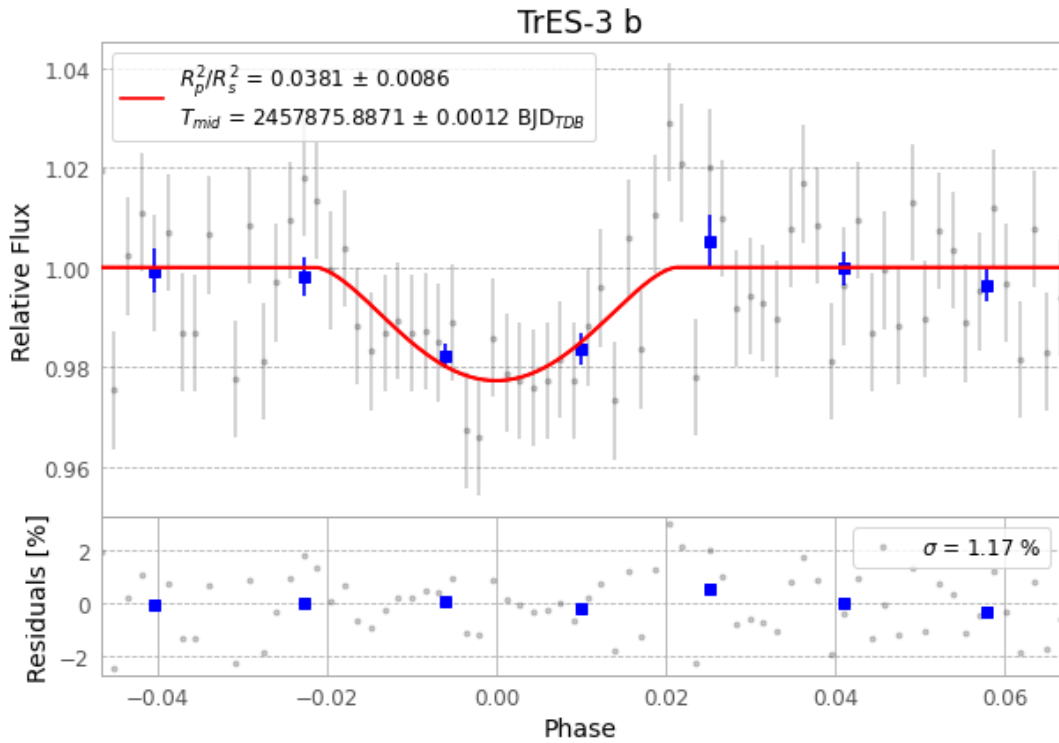


Figure 1 — FinalLightCurve_TrES-3 b_2017-05-02.png (SHA-256 0d7679462f274193...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 746b66a2c65df31c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

72 FITS frames (47 MB), TRES-3170502074512.FITS → TRES-3170502111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170502.log (cdcb3e93f4aa...), FinalLightCurve_TrES-3 b_2017-05-02.png (0d7679462f27...), FinalLightCurve_TrES-3 b_2017-05-02.csv (4555ee87510f...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 03:28Z.